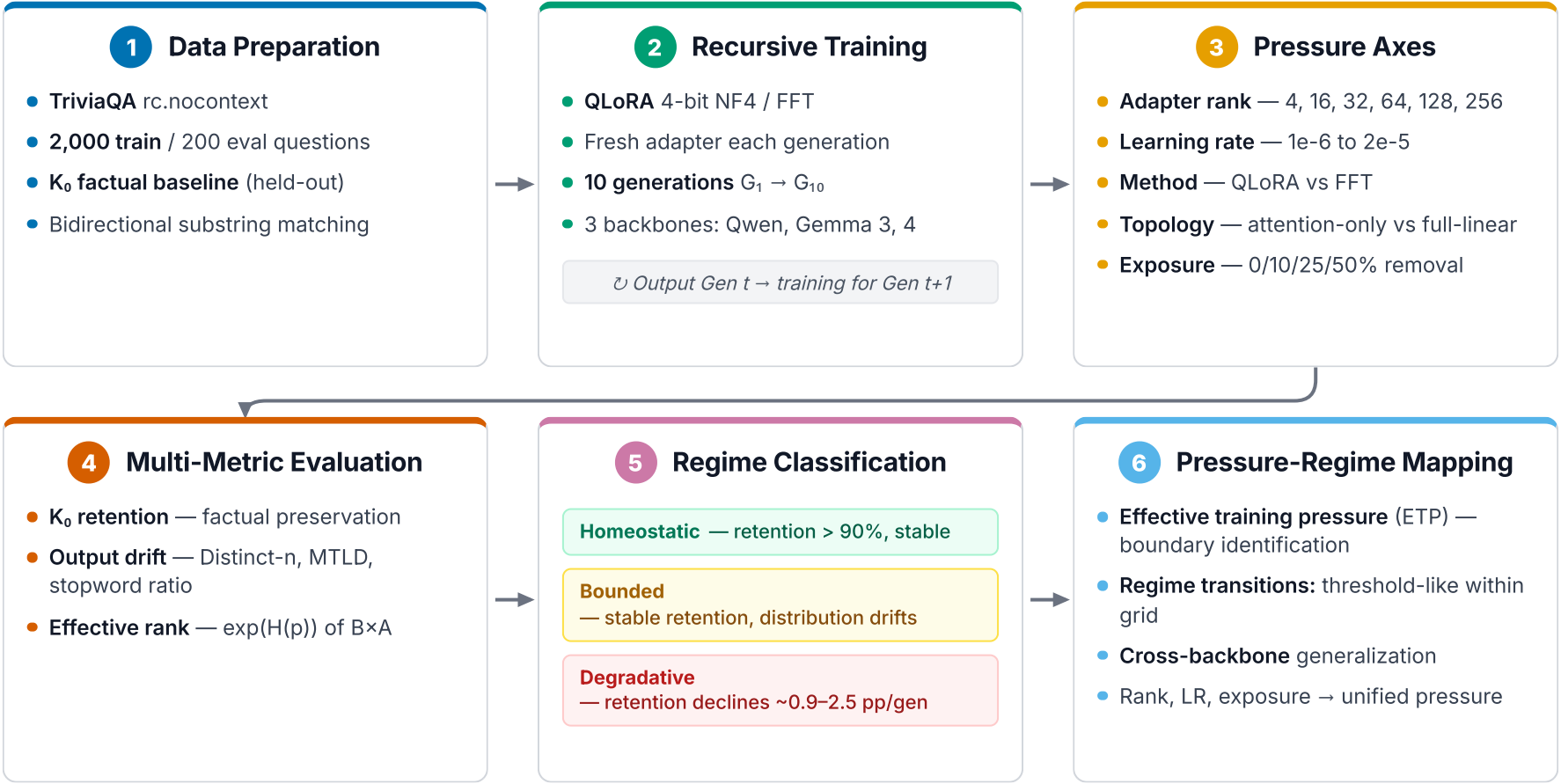

1
Data Preparation
TriviaQA rc.nocontext
2,000 train / 200 eval questions
K₀ factual baseline (held-out)
Bidirectional substring matching
2
Recursive Training
QLoRA 4-bit NF4 / FFT
Fresh adapter each generation
10 generations G₁ → G₁₀
3 backbones: Qwen, Gemma 3, 4
↻ Output Gen t → training for Gen t+1
3
Pressure Axes
Adapter rank — 4, 16, 32, 64, 128, 256
Learning rate — 1e-6 to 2e-5
Method — QLoRA vs FFT
Topology — attention-only vs full-linear
Exposure — 0/10/25/50% removal
4
Multi-Metric Evaluation
K₀ retention — factual preservation
Output drift — Distinct-n, MTLD, stopword ratio
Effective rank — exp(H(p)) of B×A
5
Regime Classification
Homeostatic — retention > 90%, stable
Bounded
— stable retention, distribution drifts
Degradative
— retention declines ~0.9–2.5 pp/gen
6
Pressure-Regime Mapping
Effective training pressure (ETP) — boundary identification
Regime transitions: threshold-like within grid
Cross-backbone generalization
Rank, LR, exposure → unified pressure